

Conditional Failure at the Root Level and the Reality-Mapping Laboratory

Research Foundation, Project Assessment, and Team Roadmap — OAL-CFTRL-PLAN-001-CLAUDE • v1.0 • Draft — Under Review

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1. Document Information

Field	Value
Title	Conditional Failure at the Root Level and the Reality-Mapping Laboratory: Research Foundation, Project Assessment, and Team Roadmap
Document ID	OAL-CFTRL-PLAN-001-CLAUDE
Version	1.0 (draft)
Date	September 23, 2026
Prepared by	Claude (Anthropic) — Primary Audit and Documentation Layer
Architect of record	Nathan Lewis II
Relationship to earlier Claude documents	Builds on OAL-CFTRL-RS-001-CLAUDE v1.0 (first research and analysis) and OAL-CFTRL-LAB-001-CLAUDE v1.0 (laboratory design). Supersedes neither. Section 7.9 records corrections to both.
Research method	Web research conducted on September 23, 2026, using primary and near-primary sources where they could be found. Verification status of every source is recorded in Section 13.
Canon status	DRAFT — UNDER REVIEW. ONESTAMP: PENDING (Architect). Nothing in this document is canon.

What this document is for. It supports the outline of the design, development, and framework for Conditional Failure at the Root Level (CFTRL) and its companion laboratory work. It states how the topic arose, preserves the Architect’s origin prompt verbatim, states the purpose and goal, assesses where the project stands, sets out where it is trying to go, reports research that bears on the design, and describes how this audit and documentation layer will help the whole team get there.

2. How This Topic Arose

The concept began on March 22, 2026, when the Architect dictated two ideas that were then documented in parallel by Perplexity and ChatGPT: **Conditional Failure at the Root Level**, and **the Department of Endings**, which is its operational arm. The central question was what happens when a root simulation in a nested stack fails, the layers below depend on it for maintenance and resource allocation, and those layers cannot reach upward to diagnose or repair it. Both documents were left in draft and neither received ONESTAMP.

On September 23, 2026, while reviewing the Master Change Log and Master Development Log, the Architect noticed that many concepts in the framework hold a definition or a name and a skeleton but lack supporting detail. Thinking through CFTRL then produced a further cluster of questions: how to tell a root structure from a non-root one, how a system could distinguish base reality from something that is not, how it could tell that a root failure has occurred, and then how to study what lies beyond such structures, whether real structures follow similar designs, whether a laboratory could be designed for this, and whether a reliable map is possible under incomplete information. The Architect asked each AI collaborator for research and analysis, and asked that failure be used as the example to learn from.

Date	Event
March 22, 2026	Architect's voice dumps; Perplexity OAL-CFTRL-NAIL-001 v0.1-PPLX and ChatGPT Development Review Dossier v1.0 produced (both draft)
September 23, 2026	Request 1: save the thought; research and analysis on root-level structures, base reality, and failure detection. Claude produced OAL-CFTRL-RS-001-CLAUDE v1.0
September 23, 2026	Request 2: how to study what lies beyond; similarity of real and modeled structures; laboratory design; reliable maps; failure as the teaching example; explicit non-assumptions. Responses received from ChatGPT, Gemini, Copilot, Perplexity, and Grok (OAL-CFTRL-RES-002 v0.1-GROK); Claude produced OAL-CFTRL-LAB-001-CLAUDE v1.0
September 23, 2026	Request 3 (clarification): the documentation must state its title, date, how the topic arose, the verbatim origin prompt, purpose, and goal; assess where the project is and where it is going; and describe how the team will get there, in PDF and DOCX. This document responds

3. Verbatim Origin Prompt

The Architect's words are reproduced without change. Each original line is shown as its own paragraph. Phrases in the non-assumptions list are conditions on the design, not incidental wording, and later documents must quote them together with their qualifying clauses.

3.1 Initial statement (September 23, 2026)

Can you save this thought for me?
I need you to gather research and analysis for a concept.

Conditional failure at the root level.

The study of root level structures, how they arise, how they fail, how they are maintained. Please include the original documents for this content that were generated.

How do you know the difference between a root level structure and a structure that is not? How can a system distinguish between base reality and something that is not? How can a system determine when a root level failure has occurred or when base reality has been compromised.

How do we study this? What about this concept works, what doesn't? What questions do we need to ask?

3.2 Expanded statement (September 23, 2026)

How can we study what lies beyond these structures? Can we model whether structures in reality follow a similar design? Or diverges in ways we haven't thought of? Is this feasible using simulations or another type of modeling? How? Can we design a laboratory environment to do this? How? Can you build a reliable map of reality? What basis of comparison do you have when you are operating on incomplete information? Use failure as an example to learn from.

What I do not assume in the design of this:

That I have the complete architecture and map of all simulations, of reality, of existence as we know it.

That we share a unified definition of what reality is.

That existence can be fully mapped, represented, comprehended in full, or that this is the aim of this laboratory.

That I aim to cause a root level compromise of simulations, of base reality, or any other structure.

That this is a how to guide to do any of those things.

It is intended to help in the design of simulation and reality level structure lifecycle design, maintenance, evolution, and improvement.

Within this design frame It does not assume that we have complete knowledge or understanding of what that is.

How This topic arose from me thinking about conditional failure at the root level The master change log and master development log. I realized that there are a lot of concepts within that framework that have a definition or a name and a skeleton attached to them. They are a rough draft, they are an incomplete outline, there are a lot of additional details and supporting information that need to be added. The goal here is to design documentation to help support, design, and develop this concept. What I will include are the responses of all of the other AI collaboration team members, where they have generated documentation, I will include the documentation, and then I will include this complete response as well.

3.3 Clarifying instruction (September 23, 2026)

My task for you is to do research to support an outline of this design, development, and framework. If that was not clarified in the beginning i am clarifying now. the documentation that you generate should have a proper title, date, how this topic arose, the verbatim origin prompt from me, purpose, and goal. it should create an assesment of where we are with this project and where we are trying to go and how you plan on helping the entire team get there. please be detailed in your research and analysis. please generate in pdf and docx. I am providing this prompt to you because some of the other ai's needed additional clarification.

4. Purpose

1. To give the team a single, research-backed foundation for the outline of the CFTRL design, development, and framework.
2. To assess honestly where the project stands: what exists, what is agreed, what conflicts, and what only the Architect can decide.
3. To state where the project is trying to go, in terms that can be checked.
4. To set out how this audit and documentation layer will help each collaborator contribute, without replacing anyone's judgment or the Architect's authority.
5. To preserve the Architect's words and the provenance of every claim so that later documents cannot drift from them.

5. Goal

5.1 Overall goal

To support the design of simulation-level and reality-level structure lifecycles (design, maintenance, evolution, and improvement) by producing documentation that lets the team study root-level structures, the limits of what can be known about what lies beyond them, and the conditions under which they fail, without assuming complete knowledge.

5.2 Working goals

#	Goal	Evidence that it has been reached
G1	A reconciled shared vocabulary for "root," "base reality," "failure," and "map"	Glossary approved by the Architect; collaborators cite it
G2	A failure taxonomy the team can use to compare runs and incidents	Taxonomy approved; used in at least one lab run and one case study
G3	A formal CFTRL specification: definition, boundary rules, and an explicit statement of the conditions	Specification reviewed; each condition testable
G4	A Department of Endings charter: placement, independence, logging, and	Charter reviewed; open questions closed or recorded

#	Goal	Evidence that it has been reached
	feedback	
G5	A minimum viable laboratory that can score inference methods against known answers	Lab run with recorded baselines and calibration
G6	A map schema with evidence tiers, provenance, and review dates	Schema adopted for all new claims
G7	Documentation maintained under version control with provenance for every claim	Change log and provenance records current

5.3 What the goal does not include

Consistent with the Architect’s non-assumptions: it does not aim at a complete map of reality or existence; it does not assume a single definition of reality; it does not aim to cause a compromise of any structure; and it is not a how-to guide. It does not ask any collaborator to decide whether base reality is simulated.

6. Where We Are: Assessment

Limits of this assessment. The Master Change Log, the Master Development Log, and the Abundance Set document (OAL-AS-001 v1.0.0) were referred to in the source material but were not available to this audit layer. Statements about them here rest on how the other documents describe them.

6.1 Inventory of documents

Document	Author	Date	Status	Contribution
OAL-CFTRL-NAIL-001 v0.1-PPLX	Perplexity	Mar 22, 2026	Draft, not canon	Original structured capture of CFTRL and the Department of Endings; explicit open questions; retracts an earlier AI-inferred definition
Development Review Dossier v1.0	ChatGPT	Mar 22, 2026	Draft, not approved	A draft “canonical” definition; testing doctrine; recommended next steps
OAL-CFTRL-RS-001-CLAUDE v1.0	Claude	Sep 23, 2026	Draft	Four senses of “root”; failure taxonomy; detection signals; 25 questions; Python model sketch
OAL-CFTRL-RES-002 v0.1-GROK	Grok	Sep 23, 2026	Draft, not canon	Five origins of roots; rights-mapping (termination, allocation, audit, seal); compromise as objective change; study protocol
OAL-CFTRL-	Claude	Sep 23,	Draft	Laboratory design (RSRML);

Document	Author	Date	Status	Contribution
LAB-001- CLAUDE v1.0		2026		evidence tiers; map schema; failure-to-map protocol
ChatGPT response	ChatGPT	Sep 23, 2026	Chat response	Root Attestation Problem; Root Identification Blind Test; detection versus attribution; separation of claims by evidentiary status
Gemini “Document 771- ALPHA”	Gemini	Sep 23, 2026	Chat response	Concise study outline and essential questions; contains an unrelated log entry (see Section 6.5)
Microsoft Copilot response	Copilot	Sep 23, 2026	Chat response	Five-test root diagnostic; control-boundary analysis; question bank including ethics and governance
Perplexity response	Perplexity	Sep 23, 2026	Chat response	Boundary-relative definition; four-way anomaly disambiguation; nine-category failure taxonomy; experiment matrix; evidence discipline

None of these carries ONESTAMP. All are drafts.

6.2 What is established (broad agreement across collaborators)

- CFTRL concerns stacked or nested systems in which lower layers depend on an upstream layer for rules, maintenance, and resources, and cannot reach upward to diagnose or repair it.
- Lower layers will tend to misread root failure as local defect. This is the central prediction and it is testable.
- Base reality cannot be proven from inside. Five of six responses say so directly or in spirit.
- “Root” is relative to a system boundary, and a stack can have more than one root.
- Correlated, time-aligned degradation across otherwise independent components suggests a shared upstream cause, but correlation alone is not proof.
- Redundancy has to be designed in before a crisis, and its independence matters more than its count.
- The Department of Endings is the operational arm of CFTRL and is meant to ensure that no failure is wasted.
- Modeling in Python is the agreed first computational step.

6.3 What is unresolved

Area	Open matter	Who can resolve it
Definition	Which sense of “root” is canonical; whether “base reality” in OAL is cosmological or an operational role	Architect
“Conditional”	The condition set is stated in prose,	Architect, with ChatGPT and

Area	Open matter	Who can resolve it
	not as a testable predicate	Claude drafting
Root versus base	Copilot and Gemini treat them as the same in places; the others separate them	Architect
Failure taxonomy	Four different taxonomies exist and are unreconciled	Team, Architect approves
Department of Endings	Placement, independence, staffing, reporting, technician profile, relation to the Antithesis Set	Architect
Ethics	Whether running every endgame is permissible if agent-like entities are involved	Architect
Python model	Sketched three times, not built; minimum viable scope not fixed	Architect chooses; team builds
Nail Prototype track	Two source conflicts remain (which variant government personnel requested; whether healing-factor testers are “preferred” or “required”)	Architect
Evidence standards	No adopted rule for what a claim needs before it enters the framework	Architect

6.4 Maturity assessment by component

Scale: **0** not started; **1** named, with a skeleton; **2** drafted independently by more than one collaborator but not reconciled; **3** reconciled and Architect-reviewed; **4** tested in the laboratory; **5** ratified.

Component	Level	Basis	Main gap
CFTRL core definition	2	ChatGPT’s draft definition; refinements by Perplexity and Grok	Not reconciled; “conditional” not formalized
Root typology and identification tests	2	Four, five, and six-part typologies across collaborators	Needs one glossary
Failure taxonomy	2	Claude (8 modes), Perplexity (9), Grok (8), Copilot (4)	Needs merge and definitions
Base-reality distinction	2	Broad agreement that it cannot be proven from inside	Copilot/Gemini divergence; evidence tiers not adopted
Detection and attribution	2	Signals lists from all collaborators	No metrics or scoring
Department of	1–2	Concept plus charter proposals	Placement, independence, and

Component	Level	Basis	Main gap
Endings			open questions
Python model	1	Three specifications, no code	Scope decision
Laboratory (RSRML)	1–2	Claude’s design; ChatGPT’s blind test; Grok’s minimal stack	Not built; no scoring yet
Map schema and evidence tiers	1–2	Claude’s schema; ChatGPT’s status classes	Not adopted
Provenance and versioning	3	Versioning records, verbatim records, and audit diffs are in regular use	Stray content and collaborator additions need consistent tagging

Overall: the project has a strong, convergent conceptual core (level 2) and no tested results (level 4). The next gains come from reconciliation and from building the smallest testable lab, not from more independent drafts.

6.5 Risks

#	Risk	Effect	Mitigation
R1	Definition drift between collaborators	Different documents study different things	Single glossary; every claim states its sense of “root”
R2	“Root” and “base reality” treated as the same	Claims about built stacks and about the cosmos blur	Keep separate; cosmology claims are tier T4
R3	Consensus illusion	Agreement among collaborators read as independent confirmation	Track overlap of sources; count agreement as weak evidence
R4	Unverified citations enter the framework	Errors become part of the record	Verification queue (Section 10)
R5	Quoting a phrase apart from its qualifying clauses	The Architect’s meaning is changed	Quote phrases with their qualifiers; keep verbatim records
R6	Stray or third-party content in compiled material	A schedule fragment naming a person appears in Gemini’s response	Confirm and remove before public archiving
R7	Lesson bloat in the Department of Endings	Reports accumulate without being tested	Lessons stored as testable claims with review dates
R8	Lab worlds unrepresentative of real ones	Methods look good in the lab only	Alien-law worlds; real-record desk
R9	Scope creep into cosmology	Time spent on questions that cannot be resolved	Boundary Register; evidence tiers

#	Risk	Effect	Mitigation
R10	Tool artifacts in quoted text	Source-link markers mistaken for content	Retain verbatim and label as artifacts
R11	Dual-use drift	Deception or compromise scenarios become operational	Observational and defensive scope; no live-system work

7. Research Findings

All findings in this section were checked against sources during this session unless marked otherwise. Section 13 gives the status of each source.

7.1 Roots of trust and the “trusting trust” problem

What the sources say. The U.S. National Institute of Standards and Technology defines roots of trust as highly reliable hardware, firmware, and software components that perform specific, critical security functions, and notes that because they are inherently trusted they must be secure by design. Another NIST definition calls a root of trust simply “a starting point that is implicitly trusted.” Ken Thompson’s Turing Award lecture, “Reflections on Trusting Trust,” was published in *Communications of the ACM*, August 1984, vol. 27, no. 8, pp. 761–763; it shows how a compiler can be altered to insert a backdoor into the login program and to reinsert itself into new copies of the compiler, so that the source code looks clean. David Wheeler’s Diverse Double-Compiling (ACSAC 2005, pp. 28–40; expanded in a 2009 dissertation) recompiles the compiler’s source with a second, trusted compiler and compares the result bit for bit to the untrusted binary. Wheeler’s own account credits an earlier Air Force evaluation of Multics (Karger and Schell, 1974) as the origin of the attack.

What it means for CFTRL.

- A root of trust is defined by being *implicitly trusted*. That makes “root” partly a trust decision and only partly a fact. CFTRL should distinguish a **declared root** (a component the design treats as trusted) from a **demonstrated root** (one tested by the identification tests in LAB-001).
- Thompson’s result is the real-world form of the “who verifies the verifier” problem: inspecting everything above the root can find nothing.
- Wheeler’s technique shows what a practical answer looks like. It does not prove anything from within. It uses an independent second implementation to expose disagreement. Its assumptions must be stated: at least one independent implementation must be trustworthy, and disagreement is a prompt to investigate, not a verdict. This is the template for the lab’s “independent witness” (Question 25 in ChatGPT’s list).

7.2 Two documented failures

7.2.1 Facebook, October 4, 2021 (from the company's own post-mortem)

According to the company's post-mortem, the trigger was a command issued during routine maintenance to assess global backbone capacity, which unintentionally took down all connections in the backbone and disconnected the data centers globally. The company's systems are designed to audit such commands, but a bug in the audit tool prevented it from stopping this one. A second effect followed: the smaller facilities that answer DNS queries withdraw their route advertisements if they cannot reach the data centers, as an indicator of an unhealthy network. With the whole backbone down, the DNS servers declared themselves unhealthy and withdrew, which made them unreachable even though they were still working. Engineers then found that their normal access routes were down and that the loss of DNS broke many internal tools. The post says primary and out-of-band network access was down, so engineers were sent on site; the facilities are built with strong physical and system security, which slowed recovery. Restoring service also carried a risk: turning everything on at once could cause a surge, with individual data centers reporting drops in power use of tens of megawatts. The company credits long-running "storm" drills for preparing it for the restart, and states it had never simulated the global backbone going offline.

Observation from the post-mortem	CFTRL reading
The audit tool that should have stopped the command had a bug	The verifier failed; a verifier is itself a dependent layer
DNS servers withdrew routes when they could not reach the data centers	A protective rule designed for local health turned a backbone loss into total external unreachability: protective-action amplification
Normal and out-of-band access were both down	The "independent" path was not independent in this failure; independence has to be tested, not assumed
Internal diagnostic tools depended on DNS	Upward inaccessibility in practice: the recovery path lay inside the failure
Security hardening slowed recovery, and the company judged that a worthwhile trade-off	Lifecycle trade-off between protection and recoverability
Turning services	Cold-start is a hazard of its own

Observation from the post-mortem	CFTRL reading
back on all at once was itself dangerous	
Drills prepared them for restart but not for the failure that occurred	A real example of an endgame not yet run

7.2.2 CrowdStrike, July 19, 2024 (from the company’s root cause analysis)

The company’s analysis describes a content update in which the sensor’s interpreter expected 20 input fields while the update, validated on the assumption of 21, referenced a 21st field. The mismatch caused an out-of-bounds memory read and a system crash. The company describes the crash as the confluence of several issues: the mismatch between the 21 inputs validated and the 20 provided, the latent out-of-bounds read in the interpreter, and the lack of a specific test for non-wildcard matching in the 21st field. Earlier deployments had used wildcard matching for that field, so the latent fault stayed dormant. Its mitigations include validation at compile time, runtime bounds checks, wider test coverage, a correction to the validator’s logic, and staged deployment.

Observation	CFTRL reading
The validator and the executor held different models of the same interface	A verifier can be wrong in a way that is consistent with itself; model disagreement between verifier and executor is a failure class
Earlier successes established only that a limited set of conditions was safe	Closed-world error: “untested” treated as “safe”
The company calls it a confluence of issues	Multiple contributing conditions rather than one root cause; matches Perplexity’s and Grok’s warnings about single-cause bias
Staged deployment was added	Limiting the reach of a bad update is a direct way to reduce coupling

What was not verified in this session: claims about the recovery effort after the CrowdStrike incident (for example, that recovery needed hands-on work at each machine). RS-001 made that claim; treat it as unverified.

7.3 Why failure can be abrupt: interdependent networks

Buldyrev, Parshani, Paul, Stanley, and Havlin (*Nature* 464:1025–1028, 2010) studied cascades in interdependent networks, taking as their motivating example the September 28, 2003 near-nationwide power cut in Italy that also brought down the internet. They report that, surprisingly, a broader degree distribution makes interdependent networks *more* vulnerable to random failure, the opposite of what happens in a single network. Related work in the same line of research describes conditions under which such networks collapse in an abrupt first-order-like transition versus failing gradually in a second-order-like transition, and a paper by Parshani, Buldyrev, and Havlin (*Physical Review Letters* 105:048701, 2010) is titled as showing that reducing the coupling strength leads to a change from a first-order to a second-order percolation transition.

What it means for CFTRL.

- It gives a candidate mechanism for the “wishbone ceiling,” where the source documents say the next event is a snap rather than a slowdown: in interdependent systems, dependence itself can create abrupt collapse. This remains a hypothesis to test in the lab (Experiment E1 in RS-001).
- It suggests a design lever. If reducing coupling changes the collapse from abrupt to gradual in model networks, then redundancy that also reduces coupling may do more than redundancy that merely adds copies. This is a claim about model networks and needs testing in the lab’s worlds.
- Two details were confirmed from a reference list and an abstract rather than from the full papers. Read the originals before relying on the specific results.

7.4 Root cause localization is already a benchmarked research field

Recent work in microservice systems studies exactly the local-versus-inherited-failure problem, using known-answer testbeds with injected faults. **RCAEval** (Pham, Zhang, Ha, Salim, and Zhang; The Web Conference 2025) provides 735 failure cases from three benchmark systems (Online Boutique, Sock Shop, and Train Ticket) covering 11 fault types, with the root-cause service and the root-cause indicator annotated for each case, plus an evaluation framework with fifteen reproducible baselines. Its authors report that current methods have considerable room for improvement in accuracy across fault types. A separate study, “Root Cause Analysis for Microservice System based on Causal Inference: How Far Are We?” (arXiv 2408.13729), evaluated causal-inference methods on synthetic datasets and on the same three benchmark systems, and notes that Train Ticket, with 64 services, has longer and more complex failure propagation paths than the smaller systems. A further paper reports evaluation on both production data from large-scale systems and the public benchmarks.

What it means for CFTRL.

- The laboratory idea has direct precedent. Known-answer worlds, blind methods, and scored baselines are how this field works.
- The precedent is narrower than the lab proposed in LAB-001. Based on what these sources describe, RCAEval labels which *service* in a known system is the root cause. The RSRML would additionally test: identification of the root itself when the dependency

map is unknown; unreliable or falsified reporting from the root; worlds with laws the analysts were not shown; calibration of stated confidence; and whether a method abstains when it should. These additions are the proposed contribution.

- This section reflects the sources retrieved in this session. It is not a claim that no other work covers these additions.
- Practical route: the lab's Real-Record Desk could use RCAEval-style benchmarks as a ready-made first test bed.

7.5 Chaos engineering as the working analogue of the Department of Endings

The published Principles of Chaos Engineering describe experiments that (1) define a measurable steady state, (2) hypothesize that it persists in both a control group and an experimental group, (3) introduce variables reflecting real-world events such as crashed servers or severed network connections, and (4) try to disprove the hypothesis by looking for a difference in steady state. The principles listed are: build a hypothesis around steady-state behavior, vary real-world events, run experiments in production, and automate experiments to run continuously. The practice was pioneered at Netflix and described in Basiri et al. (*IEEE Software*, 2016; arXiv 1702.05843). Commentary on the practice adds that experiments should minimize blast radius.

What it means for CFTRL.

- It supplies the experimental discipline the Department of Endings lacks: a stated hypothesis, a control group, and a disconfirming test.
- It highlights a real difference. Chaos engineering must limit harm to live systems, so it minimizes blast radius. The Abundance Set is described as zero-cost, which removes that constraint for simulated endgames. The consequence is that the Department can run the extreme endgames that real organizations cannot, and the Facebook post-mortem shows why that matters: a total loss of the backbone was an endgame not yet simulated.
- The difference also creates a risk. Endgames run in a zero-cost set are only informative if the simulated system is faithful to the real one. The model-to-reality gap is the Department's key weakness and the reason the Real-Record Desk exists.

7.6 The simulation and base-reality debate

A 2025 paper by Faizal, Krauss, Shabir, and Marino, "Consequences of Undecidability in Physics on the Theory of Everything" (arXiv:2507.22950, submitted July 29, 2025; published in the *Journal of Holography Applications in Physics*), argues that a fully algorithmic theory of everything is impossible and that, as a consequence, the universe cannot be a simulation. It has drawn both support and objection. One published discussion letter fully endorses its conclusions but disagrees on one point about the goal of completeness. Other commentators argue that it conflates the limits of proving things with the limits of running a process, and that a system can be computational without being a simulation running on something else.

Position for OAL. This is a contested, unresolved question, and the framework does not depend on it: as Grok's dossier also noted, the Architect's nested stacks can fail at their roots whether or not the cosmos is a simulation. Claims about it should carry evidence tier T4 in the

LAB-001 scheme. Other simulation-hypothesis references cited by collaborators (Bostrom, Chalmers, Ćirković, Torres, and Beane, Davoudi, and Savage) were not re-verified in this session.

7.7 Identifiability: what can be inferred from inside

Edward Moore’s “Gedanken-experiments on Sequential Machines” (in *Automata Studies*, edited by Shannon and McCarthy, Annals of Mathematics Studies 34, Princeton University Press, 1956, pp. 129–153) treats finite automata from the experimental point of view: what an experimenter can determine about a machine from inputs and outputs. The paper concerns strictly deterministic machines and says extending the approach to probabilistic machines is not attempted.

What it means for CFTRL.

- It is the classical basis for the identifiability scoring proposed in LAB-001 (which hidden structures are identified, partially identified, or not identifiable given an observation port).
- Its limit matters for the lab: real observation is noisy and the worlds the Architect describes may not be deterministic. Moore’s results are a starting point for deterministic worlds only. Noisy and probabilistic worlds need other tools (for example, statistical identifiability and causal inference) and should be treated as an open research task.

7.8 Consolidated findings

#	Finding	Evidence (this session)	Design consequence
F1	“Root of trust” is defined by implicit trust	NIST glossary	Separate declared roots from demonstrated roots in the glossary
F2	Verifiers fail, and can fail consistently with themselves	Facebook audit-tool bug; CrowdStrike validator-executor mismatch; Thompson	Verifier independence and model agreement as first-class requirements
F3	Protective logic can amplify a failure	Facebook DNS/BGP withdrawal rule	Add “protective-action amplification” to the taxonomy
F4	Out-of-band paths can share the failure	Facebook post-mortem	Test independence in the lab, not just declare it
F5	Hardening and recoverability trade off	Facebook post-mortem	Record the trade-off explicitly at design time
F6	Restart is a hazard of its own	Facebook post-mortem (surge risk)	Cold-start design and drills
F7	Latent faults hide behind limited test coverage	CrowdStrike RCA	Department of Endings must cover untested regions of an interface

#	Finding	Evidence (this session)	Design consequence
			contract
F8	Failures usually have several contributing conditions	CrowdStrike RCA (“confluence”)	Avoid single-root-cause reporting
F9	Interdependence can make collapse abrupt; coupling strength is a lever	Buldyrev et al.; Parshani et al. title	Candidate mechanism for the ceiling; test in lab
F10	Independent second implementations expose subversion by disagreement	Wheeler DDC	Independent witnesses; disagreement is a prompt to investigate
F11	Known-answer failure benchmarks exist and leave room for improvement	RCAEval	The lab is viable; use existing benchmarks as an early test bed
F12	Failure experiments need hypotheses, controls, and disconfirmation	Principles of Chaos Engineering	Adopt in the lab’s protocols
F13	Base-reality claims are contested	FKSM paper and replies	Tier T4; not a dependency of the framework
F14	Identifiability theory for deterministic machines is old; noisy worlds are open	Moore 1956	Deterministic worlds first; noisy worlds as a research task

7.9 Corrections to earlier documents

Research in this session found points in earlier Claude documents that should be corrected or qualified.

Document and section	Statement	Correction
RS-001, Section 6.3	Thompson’s “1984 Turing Award lecture”	The lecture was published in <i>Communications of the ACM</i> , August 1984. Sources differ on whether to call it the 1983 or 1984 award lecture. Cite the CACM publication (1984).
RS-001, Section 7.2; LAB-001, Section 7.4	The Facebook outage described as a configuration change that withdrew routes	The company’s post-mortem says a maintenance command intended to assess backbone capacity took down the backbone, an audit-tool bug failed to stop it, and DNS servers then withdrew their route advertisements as designed. Section 7.2.1 gives the fuller account.

Document and section	Statement	Correction
RS-001, Section 8.1	The 2024 CrowdStrike incident “required hands-on machine-by-machine recovery”	Not verified in this session. The cause is documented in the company’s root cause analysis (Section 7.2.2).
RS-001, Section 13; LAB-001, Section 11	Bibliographies “not verified”	Section 13 of this document records which items were verified in this session.

8. Analysis: What the Research Means for the Framework

1. **“Root” is a decision as well as a fact.** NIST’s definition makes roots of trust things the system chooses to trust. CFTRL should therefore record, for each root, whether it is declared, demonstrated by tests, or both, and by whom.
2. **The verifier is part of the stack.** Both real failures reviewed here involved a checking mechanism that did not catch the fault. The framework should never treat a check as outside the failure domain without showing that it is.
3. **The taxonomy needs at least three additions from real cases:** protective-action amplification, verifier–executor model disagreement, and cold-start hazard. These sit alongside the eight modes in RS-001, the nine categories from Perplexity, and the eight classes from Grok.
4. **The “ceiling” has a candidate mechanism.** Interdependence and coupling strength give the wishbone image a formal partner, and produce a testable prediction: reducing coupling should soften the transition.
5. **The laboratory is feasible and partly precedented.** The novel parts are the ones the Architect’s non-assumptions call for: unknown dependency maps, unreliable reporting, alien laws, calibrated uncertainty, and honest abstention.
6. **The Department of Endings needs an experimental method and a fidelity rule.** Chaos engineering supplies the method. The fidelity rule (how a simulated root is shown to resemble the real one) is the missing piece.
7. **Base reality can be handled by boundary, not by proof.** The debate is unresolved, so the framework should record such claims in the Boundary Register and proceed without them.
8. **Multiple contributing conditions are the norm.** A CFTRL report should list contributing conditions and their interactions, not name one root cause.

9. Where We Are Going

9.1 Target state

A set of reconciled, Architect-approved documents and a working minimum laboratory that together let the team:

1. speak about roots, failures, and maps using one vocabulary;
2. classify failures with one taxonomy;
3. state CFTRL as a specification whose conditions can be tested;
4. run the Department of Endings under a written charter with an experimental method;
5. score inference methods against known answers and report calibration and abstention;
6. maintain a map with evidence tiers, provenance, and review dates;
7. and learn from each failure through the failure-to-map protocol.

9.2 Roadmap

The roadmap is an ordered sequence with gates. Timing is the Architect's to set. Each gate ends with an Architect decision.

Phase	Work	Main deliverables	Gate (exit criterion)
0. Reconcile	Merge vocabularies and taxonomies; resolve the root/base distinction; settle open conflicts; clean the compiled material; run a provenance diff across all responses	Glossary v0.1; taxonomy v0.1; provenance diff; decisions D1–D8	Architect approves glossary and taxonomy
1. Specify	Write the CFTRL formal specification; the Department of Endings charter; the minimum lab specification; the map-entry schema	CFTRL specification v0.2; DoE charter v0.1; MVP lab spec; map schema	Architect reviews each; open questions closed or recorded
2. Build	Build the minimum viable lab (chain of four layers; three failure modes; baselines; one method; scoring)	Working code; scoring report format	Runs end to end; baseline scores recorded
3. Run and score	Blind mapping across single-factor failures; calibration and abstention measured	First lab report; Misfit Ledger opened	At least one method beats baselines with calibrated confidence, or a negative result is reported
4. Extend	Unreliable-reporting worlds; alien-law worlds; Real-Record Desk using public benchmarks and post-mortems	Second lab report; case-study records using the failure-to-map protocol	Map-error rate estimated with stated bias
5. Integrate	Feed results into lifecycle documents; update the master logs; put Department of Endings procedures into practice	Lifecycle mapping; updated logs	Architect confirms integration

Phase	Work	Main deliverables	Gate (exit criterion)
6. Ratify	Architect review, corrections, ONESTAMP where approved; archive with versioning	Ratified documents	ONESTAMP applied by the Architect

9.3 What “done” looks like for each deliverable

Deliverable	Done when
Glossary	Every term used by two or more documents has one approved definition or a recorded reason for two
Taxonomy	Every failure mode has a definition, an observable signature, and an example
CFTRL specification	Each condition in “conditional” is stated as a testable predicate
Department of Endings charter	Placement, independence, logging format, reporting line, and feedback path are stated
Minimum lab	Reproducible run with baselines, scores, and calibration
Map schema	Adopted for all new claims, with evidence tier and review date
Provenance record	Every claim in a ratified document traces to a source and an author

10. How This Layer Will Help the Whole Team

A limit to state plainly. This layer works only within conversations the Architect opens. It cannot contact the other collaborators. Its help reaches them through documents and briefs that the Architect relays.

10.1 What Claude will do

Function	What it involves
Provenance and versioning	Versioning records; verbatim source records; tagging each claim by author and origin, including collaborator additions
Reconciliation	Provenance diffs across parallel drafts; a merged list of terms, taxonomies, and questions with sources
Verification	A queue of unverified citations, checked against primary sources, with the status recorded
Audit	Review of claims for definition drift, single-cause bias, consensus illusion, and unsupported leaps; findings reported, never enforced
Drafting support	First drafts of the glossary, taxonomy, specification skeleton, charter skeleton, map schema, and lab protocol for the Architect’s decision
Lab protocol	Scoring rules, independence rules, and pre-registration format for the minimum lab
Clarity for collaborators	A paste-ready task brief (Appendix A) so each collaborator receives the same context, wording, and constraints

Function	What it involves
Decision support	Decisions D1–D8 stated as concrete choices with consequences, so the Architect can decide quickly

Claude will not adjudicate disputes between collaborators, decide canon, or apply ONESTAMP. These belong to the Architect and the existing OAL governance processes.

10.2 Proposed contributions from each collaborator

These are proposals for the Architect to accept or change. They follow what each response has already shown strength in.

Collaborator	Proposed contribution	Why
ChatGPT	Formal specification of CFTRL; Root Attestation Problem architectures (single, dual, cross-attesting, quorum, external watchdog); Root Identification Blind Test protocol	Its response gave the most complete formal framing
Perplexity	Literature and source verification; mapping known-answer benchmarks and post-mortems to the Real-Record Desk; the evidence-discipline checklist	Its role in the workflow is research, and its response gave the boundary-relative definition and evidence standards
Grok	Rights-mapping (termination, allocation, audit, seal) and objective-change detection; the first Python minimum stack, if the Architect chooses	Its dossier gave the most operational tests and a concrete minimum build
Microsoft Copilot	Department of Endings charter: reporting lines, obligations to dependent layers, and how findings avoid being ignored; the ethics question	Its response raised governance and ethics most directly
Gemini	Consolidated question bank; confirmation of the stray log entry and its source; maintaining the Architect's archive rules	Its response gave a concise question list; the stray entry needs its attention
Claude	Provenance, reconciliation, verification, audit, laboratory protocol and scoring	Its established lane

10.3 Working agreements proposed for the team

1. **Nothing is canon until ONESTAMP.** Every document says so.

2. **Every claim has a source and a tier.** Use the tiers T1 to T4 from LAB-001, or a variant the Architect prefers.
3. **Quote with qualifiers.** A phrase from the Architect is quoted together with its qualifying clauses (Risk R5).
4. **Sense of “root” is always stated.**
5. **Unverified means unverified.** Citations from memory or from another collaborator are labeled until checked.
6. **Disagreements are recorded, not smoothed.** Conflicts go to the Architect with the options laid out.
7. **Agreement among collaborators is weak evidence** because sources overlap.
8. **Observational and defensive scope only.** No live-system probing and no methods to breach any structure.
9. **Version everything.** Each document carries a versioning record with origin, purpose, hypothesis, date, and preparer.
10. **Provenance of additions.** Ideas added by a collaborator that are not in the Architect’s words are tagged as collaborator-origin.

10.4 Working cycle

1. **Intake:** the Architect issues a task with the standard brief.
2. **Parallel drafts:** each collaborator drafts in its lane.
3. **Audit diff:** Claude compares drafts, flags conflicts, verification needs, and provenance.
4. **Decisions:** the Architect resolves conflicts and approves or revises.
5. **Integrate:** approved material enters the working documents with version increments.
6. **Ratify:** the Architect applies ONESTAMP where approved.

10.5 Near-term work for Claude

#	Task	Output
1	Verification pass on the citations that were not checked in this session (Section 13)	Updated source-status table
2	Draft Glossary v0.1, merging the typologies of all collaborators	Glossary for Architect review
3	Draft Failure Taxonomy v0.1, merging the four taxonomies and adding F3, F2, and F6	Taxonomy for Architect review
4	Provenance diff of all September 23 responses, tagging collaborator additions	Provenance record
5	Draft the minimum lab protocol in full (worlds, ports, scoring, pre-registration)	Protocol document
6	On instruction, build the minimum lab in Python	Code and first report

11. Decisions Needed from the Architect

#	Decision	Options	Consequence
D1	Which sense of “root” is canonical for OAL?	Dependency, authority, trust, ontological, relative-root formalism, or a stated combination	Fixes what CFTRL studies and how failures are classified
D2	Is “base reality” in OAL a cosmological claim or an operational role (the layer with no authorized parent)?	Cosmological / operational / both, kept separate	Decides whether root/base can be conflated in any document
D3	How is “conditional” to be formalized?	Structural predicate; conditional probability; both	Sets what the formal specification must contain
D4	Where does the Department of Endings sit relative to the stack it studies?	Inside, outside, or across	Sets independence and audit needs
D5	Who reviews the laboratory and the Department of Endings?	Existing OAL governance; Governance Authority Laboratory; other	Sets accountability
D6	May the lab run endgames on agent-like entities inside lab worlds?	Yes with rules; no; only for defined agent classes	Sets ethics scope
D7	What is the minimum viable lab, and who builds it?	Scope from Section 5.8 of LAB-001; builder chosen by the Architect	Starts Phase 2
D8	How are collaborator additions to be tagged and used?	Tag and hold until approved; tag and use provisionally	Sets how fast material moves toward canon

12. Limits of This Research

- Searches were run on September 23, 2026. Where a primary source was available (the Facebook engineering post-mortem, CrowdStrike’s root cause analysis, NIST’s glossary, arXiv, the ACSAC and CACM records), it was preferred. In some cases only abstracts, reference lists, or secondary summaries could be seen; Section 13 says which.
- Several items in earlier documents and in collaborator responses were not verified (Section 13).
- The Master Change Log, Master Development Log, and the Abundance Set document were not available to this layer.
- The comparison with RCAEval reflects the sources retrieved. A broader literature search might find further overlap.

- This document reports findings and proposals. It does not decide anything for the Architect.

13. Sources and Verification Status

Status key. *Verified* means the source was retrieved in this session and the stated points were confirmed. *Partly verified* means only an abstract, a reference entry, or a secondary description was seen. *Not re-verified* means the item is well established or was cited by another collaborator but was not checked in this session.

Source	Used for	Status
NIST Computer Security Resource Center glossary, “roots of trust” (csrc.nist.gov/glossary/term/roots_of_trust); NIST SP 800-172 and NISTIR 8320	Definitions of root of trust	Verified
K. Thompson, “Reflections on Trusting Trust,” <i>Communications of the ACM</i> 27(8), Aug 1984, pp. 761–763 (accessible copy at inst.eecs.berkeley.edu/~cs161/fa08/papers/p761-thompson.pdf)	Compiler backdoor; trust in the foundation	Verified (publication details; award-year labelling varies by source)
D. A. Wheeler, “Countering Trusting Trust through Diverse Double-Compiling,” ACSAC 2005, pp. 28–40 (arXiv:1004.5548; acsac.org/2005/abstracts/47.html ; dwheeler.com/trusting-trust/ ; dissertation 2009)	Diverse double-compiling	Verified
S. Janardhan, “More details about the October 4 outage,” Engineering at Meta, Oct 5, 2021 (engineering.fb.com/2021/10/05/net-working-traffic/outage-details/)	Facebook case	Verified (primary source)
CrowdStrike, “Channel File 291 Incident Root Cause Analysis,” Aug 6, 2024 (crowdstrike.com/wp-content/uploads/2024/08/Channel-File-291-Incident-Root-Cause-Analysis-08.06.2024.pdf)	CrowdStrike case	Verified (primary source, excerpts)
S. V. Buldyrev, R. Parshani, G. Paul, H. E. Stanley, S. Havlin, “Catastrophic cascade of failures in interdependent networks,” <i>Nature</i> 464:1025–1028	Interdependent-network cascades	Partly verified (abstract and citations)

Source	Used for	Status
(2010), doi:10.1038/nature08932		
R. Parshani, S. V. Buldyrev, S. Havlin, <i>Phys. Rev. Lett.</i> 105:048701 (2010)	Coupling strength and transition order	Partly verified (title from reference lists)
L. Pham, H. Zhang, H. Ha, F. Salim, X. Zhang, “RCAEval: A Benchmark for Root Cause Analysis of Microservice Systems with Telemetry Data,” WWW ’25 (arXiv:2412.17015; Zenodo record 14504480)	Known-answer failure benchmark	Verified
“Root Cause Analysis for Microservice System based on Causal Inference: How Far Are We?” (arXiv:2408.13729)	Causal-inference RCA evaluation	Partly verified (setup section)
Principles of Chaos Engineering (principlesofchaos.org, last update March 2019); A. Basiri et al., “Chaos Engineering,” <i>IEEE Software</i> (2016), arXiv:1702.05843	Experimental method	Verified
M. Faizal, L. M. Krauss, A. Shabir, F. Marino, “Consequences of Undecidability in Physics on the Theory of Everything,” arXiv:2507.22950 (2025), <i>J. Holography Applications in Physics</i> ; published discussion letters and online replies	Simulation-debate status	Verified that the paper and replies exist; content of replies summarized from abstracts and commentary
E. F. Moore, “Gedanken-experiments on sequential machines,” in <i>Automata Studies</i> (Annals of Mathematics Studies 34), Princeton UP, 1956, pp. 129–153	Identifiability of deterministic machines	Verified (bibliographic details; scope from a secondary citation of the introduction)
Kalman (1960) on observability; Pearl, <i>Causality</i> ; Milner and Park on bisimulation	Identifiability and causal inference	Not re-verified
Talts et al. (2018) on simulation-based calibration; Cranmer, Brehmer, and Louppe (PNAS 2020)	Lab calibration methods	Not re-verified
Clauset, Shalizi, and Newman (SIAM Review, 2009)	Cautions on power-law claims	Not re-verified
H. Simon, “The Architecture of Complexity” (1962); Allen and Starr (1982)	Hierarchy and timescale separation	Not re-verified

Source	Used for	Status
S. Dekker, <i>Drift into Failure</i> (2011); D. Vaughan, <i>The Challenger Launch Decision</i> (1996); C. Perrow (1984); R. Cook (1998)	Failure and normalization of deviance	Not re-verified
E. W. Dijkstra (1974) on self-stabilizing systems (cited by ChatGPT)	Self-stabilization	Not re-verified
Bostrom (2003); Chalmers, <i>Reality+</i> (2022); Beane, Davoudi, Savage (2012/2014); Ćirković; Torres	Simulation hypothesis	Not re-verified
Google SRE writing on cascading failures (cited by Perplexity and Grok)	Cascading failure	Not re-verified

Appendix A — Collaborator Task Brief (paste-ready)

Purpose: several collaborators needed additional clarification. This brief gives each one the same context, wording, and constraints. The Architect may edit it.

TITLE: Conditional Failure at the Root Level and the Reality-Mapping Laboratory — [name of the task assigned]

DATE: [date of issue]

HOW THIS TOPIC AROSE: The Architect reviewed the Master Change Log and Master Development Log and noticed that many concepts hold a definition or a name and a skeleton but lack supporting detail. Thinking about Conditional Failure at the Root Level (CFTRL) led to questions about root-level structures, base reality, failure detection, and how to study what lies beyond them. The original documents are the March 22, 2026 Perplexity document (OAL-CFTRL-NAIL-001 v0.1-PPLX) and ChatGPT dossier (Development Review Dossier v1.0).

VERBATIM ORIGIN PROMPT: [Paste Sections 3.1, 3.2, and 3.3 of this document without changes.]

PURPOSE: To support the design, development, and framework of CFTRL and its laboratory by producing research and analysis that is detailed, sourced, and honest about what is and is not known.

GOAL: Documentation that helps the team design and maintain structure lifecycles (design, maintenance, evolution, and improvement) without assuming complete knowledge of reality.

YOUR TASK: [state the specific task, for example, “draft the formal CFTRL specification” or “verify the citations in Section 13”]

DELIVERABLE FORMAT: PDF and DOCX, with a versioning record (document ID, version, date, preparer, topic origin, purpose, hypothesis), the verbatim origin prompt, and a statement of canon status (draft, ONESTAMP pending).

CONSTRAINTS (from the Architect’s non-assumptions):

1. Do not assume complete knowledge of the architecture or map of simulations, reality, or existence.
2. Do not assume a single shared definition of reality. State which sense of “root” or “reality” you use.
3. Do not aim to map everything.
4. Do not produce methods to compromise any simulation, base reality, or other structure. This is not a how-to guide.
5. Quote the Architect’s phrases together with their qualifying clauses.
6. Label anything not verified as unverified, and tag ideas that are your own additions.
7. Do not treat any draft as canon. Only the Architect applies ONESTAMP.

WHAT TO INCLUDE: an assessment of where the work stands, where it is heading, what works and what does not, the questions still open, and how your contribution helps the team.

WHAT TO FLAG: conflicts between documents, unverifiable claims, and anything in the material that appears to come from another context.

Appendix B — Change Log for This Document

Version	Date	Change	By
1.0	September 23, 2026	First issue. Research verified in-session; corrections to RS-001 and LAB-001 recorded	Claude